

What actually moves virtual screening performance: two measured resolution limits and a pre-registered decomposition of six interventions

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PREPRINT — 6 September 2026

Abstract

Reported improvements to structure-based virtual screening are usually differences in AUROC or enrichment between two protocols on one benchmark. We asked what a benchmark of typical size can actually resolve, and then measured six interventions against that limit with the reading rule for each committed to version control before the data existed.

Two limits, not one. Scoring the *same* Vina poses with a second implementation of the same scoring function — agreeing at $r = 0.970$, no compound differing by more than 2 kcal/mol — moves AUROC by 0.020 [95% CI 0.011, 0.029]. Re-running the whole protocol at a different search effort and scoring it the same way moves AUROC by 0.039 [0.024, 0.056]. The first bounds a comparison of scoring functions on fixed poses; the second bounds a comparison of protocols, which is what most published improvements are. Both fall inside the range in which docking improvements are typically reported.

On SARS-CoV-2 Mpro: changing the scoring function on identical poses moved AUROC +0.140, repairing a receptor preparation defect +0.045, scoring the pose ensemble rather than the top pose +0.019, substituting DiffDock-L pose generation +0.017, supplying the correct binding site to a co-folding model -0.013, and eight-fold search effort -0.019. Read against each arm’s own pre-registered bar, one is supported at full strictness, one directionally, and the rest are not. Read against the resolution limits measured here — which were fixed after four of the six bars, and so are reported alongside those readings rather than in place of them — one clears, one is marginal, and four fall inside. The search-effort result is the sharpest either way: its measured effect is smaller than the run-to-run reproducibility of the protocol it was measured in.

Docking adds nothing demonstrable over seven free physicochemical descriptors on either panel we assembled: on Mpro it significantly subtracts (-0.002 [-0.004, -0.000]) and on Factor Xa its +0.015 [-0.003, +0.034] crosses zero. Applying the same descriptors to LIT-PCBA gives a median AUROC of 0.7260 across 15 targets against published AutoDock Vina at 0.581 and GNINA at 0.611-0.616. These descriptor baselines are supervised on each benchmark’s own labels while docking is not, so this is a bias control in the sense of Sieg et al., not a claim that descriptors are a deployable substitute. On the AVE-debiased variant the descriptor median falls to 0.6510.

One method exceeded the floor. Boltz-2, an open-weights co-folding model scoring compounds from sequence and SMILES alone, does not beat the descriptor baseline on its own (Mpro +0.026, Factor Xa +0.019, both intervals spanning zero) but adds +0.093 [+0.062, +0.125] and +0.075 [+0.051, +0.099] respectively when combined with those descriptors. Its advantage is not that it is less property-driven — descriptors explain 63% of its score and 63% of Vina’s — but that on Mpro its residual, the part orthogonal to those descriptors, ranks actives at 0.642 where five docking scores we measured sat between 0.507 and 0.559.

1. Introduction

A virtual screening method is normally shown to work by reporting enrichment on a retrospective benchmark. Two properties of that evidence are rarely reported alongside it. The first is what difference the benchmark could distinguish from noise. The second is which component of a multi-part protocol produced the difference — absent that, a pipeline change gets attributed to whichever part its authors were working on.

Analogue and decoy bias in retrospective sets is well documented [1,2], and LIT-PCBA [3] was constructed to control it — though LIT-PCBA has itself since been audited and found to carry leakage and redundancy across its splits [8]. That literature establishes that reported enrichment can measure property matching. It does not establish how much of a *given* reported improvement is resolvable at all, which is the question here.

This paper reports a resolution limit, measured rather than assumed and in two variants; a set of six single-factor comparisons against it, each with its reading rule fixed in advance; and a descriptor baseline on LIT-PCBA, to test whether our own panels are unrepresentative.

2. Methods

Benchmarks. Two panels assembled from ChEMBL with measured actives and measured inactives: SARS-CoV-2 Mpro (PDB 7VU6, 751 compounds, 34.2% active) and Factor Xa (886 compounds, 45.7% active; an earlier 854-compound freeze of the same panel is used for the marginal-value comparison in §3.4 and is labelled where quoted). LIT-PCBA [3] full and AVE-debiased variants, 15 targets, used as distributed.

Descriptor baseline. Molecular weight, clogP, hydrogen-bond donors, hydrogen-bond acceptors, rotatable bonds, topological polar surface area, formal charge. Logistic regression, out-of-fold predictions under 5-fold stratified cross-validation. On the AVE variant the authors’ own train/validation split is honoured rather than re-folded, so the AVE and full-set figures differ in estimator as well as in compound selection. Inactives capped at 25,000 per LIT-PCBA target by seeded random sampling.

Docking. AutoDock Vina 1.2.5, 22 Å box, receptor passing a redocking control at the screening protocol. Exhaustiveness was 4 for the scoring-function, receptor-repair and pose-ensemble arms; the search-effort arm compares exhaustiveness 4 against 32. gnina 1.3 CNN rescoring applied to the identical Vina output poses. DiffDock-L for the pose-source arm, with poses minimised before scoring. Supplementary Table S2 gives the receptor state, exhaustiveness and baseline for each arm.

Boltz-2. Version 2.2.1, protein sequence plus ligand SMILES, `affinity_probability_binary` as the endpoint, fixed in advance. `diffusion_samples` = 1 for cost. Pocket conditioning, where applied, supplied as 1-based sequence indices derived from residues within 8 Å of the docking box centre.

Pre-registration. Every reading rule — endpoint, threshold, control, abort condition — was committed to a public git repository before the corresponding data existed. Supplementary Table S1 lists each arm’s pre-registration file, commit hash and commit date. Amendments are appended rather than edited, with the reason recorded. Two comparisons were not pre-registered and are identified as such in S1.

Statistics. Paired bootstrap over compounds, 95% percentile intervals: 10,000 resamples for the interventions in §3.2 and for the resolution floors, 4,000 for the Boltz-2 contrasts. A difference is reported as demonstrated only when it exceeds the applicable resolution floor *and* its interval excludes zero.

3. Results

3.1 Two resolution limits, 0.020 and 0.039 AUROC

gnina’s --score_only mode re-evaluates the Vina scoring function on a pose file it is handed. Applied to the exact poses Vina wrote at exhaustiveness 4, it reproduces the cached Vina score at $r = 0.9700$, mean difference -0.108 kcal/mol, with no compound differing by more than 2 kcal/mol. The two score vectors nonetheless give AUROC 0.3992 and 0.3791 — **0.020 apart [+0.011, +0.029]**. Call this the **scoring floor**: what two implementations of one scoring function differ by when placement is held fixed.

Comparing instead against the exhaustiveness-32 cache — same compounds, same scoring function, an independent stochastic search — gives $r = 0.9372$ and AUROC 0.4186 against 0.3793, **0.039 apart [+0.024, +0.056]**. Call this the **protocol floor**: run-to-run reproducibility of the pipeline end to end.

Neither pair is a meaningfully different description of binding, yet the metric separates each by an amount comparable to published improvements. We report a difference as unresolvable when it falls below the floor that matches the comparison being made: the scoring floor where poses are held fixed, the protocol floor where placement or search is allowed to vary.

3.2 One intervention clears its floor; four fall inside it

intervention	Δ AUROC	95% CI	pre-registered bar	pre-registered verdict
Scoring function, identical poses (gnina CNN)	+0.140	[+0.091, +0.187]	CI entirely above +0.10	directionally supported, not at full strictness (lower bound 0.091)
Receptor preparation repair	+0.045	[+0.024, +0.067]	CI entirely above 0	supported
Pose ensemble rather than top pose	+0.019	[−0.017, +0.055]	> +0.04, CI excluding 0	refuted
Pose generator substituted (DiffDock-L)	+0.017	[−0.027, +0.064]	CI above +0.05 would refute “scoring is the problem”	scoring hypothesis retained
Binding site supplied to Boltz-2	−0.013	[−0.026, +0.000]	> +0.04, CI excluding 0	not demonstrated
Eight-fold search effort	−0.019	[−0.032, −0.006]	<i>not pre-registered</i>	—

On re-gating, and why we do not do it silently. Four of these bars were fixed before the resolution limits in §3.1 were separated, and three of them cite a single “ ≈ 0.04 measurement floor” that we now know conflates the scoring and protocol cases. A pre-registration cannot be moved after the fact, so the verdicts above are read against the bars as written. The corrected floors are reported alongside as a secondary observation, and they change the picture in one place only:

intervention	poses fixed?	applicable floor	reading against it
Scoring function (gnina CNN)	yes	scoring, 0.020	clears

intervention	poses fixed?	applicable floor	reading against it
Receptor preparation repair	receptor only; ligands re-docked	protocol, 0.039	marginal — the point estimate exceeds the floor but the two intervals overlap across nearly their whole width
Pose ensemble	yes	scoring, 0.020	inside
Pose generator (DiffDock-L)	no	protocol, 0.039	inside
Binding site to Boltz-2	n/a — different model	protocol, 0.039	inside
Eight-fold search effort	no	protocol, 0.039	inside

The receptor-repair row is the one place the two readings diverge: it passes its pre-registered bar cleanly and is only marginal against a floor fixed afterwards. We report both and claim neither over the other.

These are six single-factor comparisons, not a partition of one pipeline. Four were run on the donor-defective receptor described below, before it was repaired; the receptor-repair row measures what that condition costs, and Supplementary Table S2 gives each arm’s baseline in full. Their nulls are therefore conditional on a receptor now known to be broken.

The one intervention that clears its floor outright concerns *what is computed*, not *where or how thoroughly the ligand is placed*. The search-effort row makes the point most economically: its CI excludes zero, so the effect is real and negative, but it is smaller than the protocol floor — and the exhaustiveness-4/32 pair is the protocol floor, so eight times the compute is by construction indistinguishable from running the same protocol twice. The pose-source result is next sharpest: Vina and DiffDock poses correlate at $r = 0.139$, nearly uncorrelated coordinates, and produce rankings differing by less than the floor.

The receptor defect is worth reporting for its own sake. Every receptor we had prepared carried zero hydrogen-bond donors: the converter typed all 754 Mpro nitrogens as acceptors and added no polar hydrogens, so the scoring function could not form a protein-donor hydrogen bond at all. Repairing it moved AUROC +0.045 and changed six of the top ten shortlisted compounds. A defect invisible in every summary statistic reordered most of the answer.

3.3 Seven descriptors reach 0.726 on LIT-PCBA, against published docking at 0.58-0.62

	median AUROC	targets
descriptors, LIT-PCBA full	0.7260	15
descriptors, full, ≥ 50 actives	0.6830	10
descriptors, AVE-debiased	0.6510	15
AutoDock Vina, published [5]	0.581	15
GNINA, published [5]	0.611-0.616	15

Per-target values are in Supplementary Table S4. Both published figures come from one source, Sunseri & Koes [5], Table 2; we are not aware of an independent second measurement of Vina on this benchmark. An independent 2026 benchmarking effort on LIT-PCBA [4] reports ROC-AUC “hovering near 0.6” for its strongest classical scorer, but its docking arm is AutoDock-GPU rather than Vina, so it

is not a substitute for the Vina figure. On the full set, seven descriptors costing microseconds exceed published Vina by 0.145, or 0.102 restricting to the ten targets with ≥ 50 actives. Our own panels’ descriptor baselines (Mpro 0.7654, Factor Xa 0.7041) fall inside LIT-PCBA’s 0.577–0.864 range, so they are not unrepresentative.

What this comparison is and is not. Our descriptor baselines are supervised on each target’s own actives and inactives; docking scores require no labels. The comparison is therefore a bias control in the sense of Sieg et al. [2] — it bounds how much of a benchmark’s apparent enrichment is reachable from ligand properties alone — not a claim that descriptors are a deployable substitute for docking on a novel target. Sunseri & Koes [5] reach the opposite conclusion, that their CNN models significantly outperform simple-descriptor models; their descriptor models were fit off-benchmark to PDBbind and CrossDocked affinity data and transferred, and the difference in conclusion is a difference in what the baseline is allowed to see.

What was not matched. Our baselines use up to 25,000 inactives per target drawn by seeded random sampling, against published figures computed on the full inactive sets and, in [5], taking the maximum score over all reference receptors for the 13 of 15 targets with multiple templates. AUROC is insensitive to negative-class size in expectation, so we expect no systematic direction to the sampling difference, but this is not a matched comparison and we do not claim one.

AVE debiasing reduces the descriptor median to 0.6510, robust to excluding targets with few validation actives (0.6510, 0.6497, 0.6510 at ≥ 0 , ≥ 20 , ≥ 50 actives). Two things change between 0.7260 and 0.6510 — the compound selection and, because the AVE variant ships its own train/validation split, the estimator — so the 0.075 drop is attributable to the AVE variant as a whole rather than to debiasing alone. What survives is that 0.651 from seven trivial properties on a deliberately debiased set is well clear of chance.

A comparison we cannot make. No published Vina number exists for the AVE-debiased set. Comparing our AVE descriptor result against a full-set docking result would not be like for like and we do not do so. Vina would presumably also fall under debiasing; by how much is unknown to us.

3.4 A co-folding model exceeds the floor, but only in combination

	Mpro	Factor Xa
descriptors	0.7654	0.7041
Boltz-2 alone	0.7913	0.7227
descriptors + Boltz-2	0.8588	0.7791
Boltz-2 – descriptors	+0.026 [−0.023, +0.074]	+0.019 [−0.030, +0.066]
combination – descriptors	+0.093 [+0.062, +0.125]	+0.075 [+0.051, +0.099]

Both halves replicate across a viral protease and a coagulation factor with independently assembled panels: the standalone comparison fails on both, the combination clears on both with overlapping intervals.

Boltz-2 is not less property-driven than docking — descriptors explain 63.1% of its score and 62.8% of Vina’s. The difference is the residual. After regressing out the seven descriptors out-of-fold, Boltz-2’s remainder ranks actives at **0.6418** on Mpro; Vina’s at 0.5382. Across five docking scores we residualised — Vina raw and repaired, two gnina heads, and Vina on Factor Xa — the remainder fell between 0.507 and 0.559. Two residualisation procedures were used across those five (out-of-fold

logistic and out-of-fold gradient-boosted), so the band is indicative rather than a like-for-like interval; Boltz-2's 0.6418 sits outside it under either. The residual analysis is single-target: we did not compute a Factor Xa Boltz-2 residual.

Residualising against ligand descriptors is a weaker test than receptor ablation. Binding free energy genuinely covaries with size and lipophilicity, so a physically correct scorer would also lose signal under this operation — that is why ligand efficiency exists. The defensible claim is narrow: the part of these docking scores orthogonal to seven descriptors does not rank actives in these sets, and on these two targets the ceiling appears to be a property of the scoring approach rather than of the task. A third scoring approach clearing it would be the test that generalises this.

4. Limitations

Two self-assembled targets for the six comparisons, and one of them (Mpro) is a target where our docking protocol scores below chance. LIT-PCBA broadens the descriptor result to 15 targets but not the comparisons, which remain a single-benchmark measurement.

Four of the six arms sit on the donor-defective receptor, as set out in §3.2 and Supplementary Table S2. Whether the pose-ensemble and DiffDock nulls survive on a repaired receptor is untested.

Four of the six pre-registered bars predate the floor correction. Three of them cite a single “ ≈ 0.04 measurement floor” that §3.1 shows was conflating two quantities. The bars stand as written and the verdicts in §3.2 are read against them; the corrected floors are reported beside those verdicts, never substituted for them.

The resolution floors are measured on one benchmark of one size, on the unprotonated receptor. They should be re-measured rather than assumed elsewhere; the method costs one extra scoring pass. An earlier version of this analysis reported a single 0.039 floor and described it as a comparison of identical poses; it is not, and the two variants are separated here.

The descriptor baselines are supervised and docking is not, as stated in §3.3. This is the limitation most likely to be mistaken for a stronger claim than we make.

No Vina number on AVE-debiased LIT-PCBA, and only one source for the full-set figure.

The Boltz-2 evaluation is contested territory. A 2025 and a 2026 paper disagree about its virtual screening performance [6,7]. Our contribution is the pre-registration and the residual decomposition, not the discovery that it works.

A caveat we withdrew. Boltz-2's first run was blind while its docking comparator was given the binding site, and we recorded that as a handicap making +0.093 a lower bound. Supplying the pocket produced 0.7783 against the blind run's 0.7913, with the two runs correlating at $r = 0.952$. The caveat was unnecessary and is withdrawn. It is reported here because a pre-registered caveat that turns out to be wrong is part of the record.

5. Data and code availability

All pre-registration files, amendments, analysis code and raw outputs are public at <https://github.com/AegisMindApp/screening-decomposition>, with the commit hash and date for each pre-registration given in Supplementary Table S1. The history is intact rather than a snapshot, so any cited commit can be resolved and dated directly (`git log -1 --format=%ad --date=iso <commit>`).

The dates are git metadata carried through from the repository this was exported from; they are not a third-party timestamp, and a reader can confirm ordering and internal consistency but cannot from this alone exclude authored metadata. The resolution-floor analysis is reproducible from that repository with no dependencies beyond the Python standard library, and asserts a positive control against previously published values before reporting; that control failed once, on 6 September 2026, and the defect it caught is recorded in the repository alongside the result.

Two comparisons — the search-effort arm and the LIT-PCBA descriptor baseline — were not pre-registered and are identified as such in S1.

References

- [1] Chen L, Cruz A, Ramsey S, Dickson CJ, et al. (2019) Hidden bias in the DUD-E dataset leads to misleading performance of deep learning in structure-based virtual screening. *PLOS ONE* 14:e0220113. doi:10.1371/journal.pone.0220113 [2] Sieg J, Flachsenberg F, Rarey M (2019) In Need of Bias Control: Evaluating Chemical Data for Machine Learning in Structure-Based Virtual Screening. *J Chem Inf Model* 59:947–961. doi:10.1021/acs.jcim.8b00712 [3] Tran-Nguyen VK, Jacquemard C, Rognan D (2020) LIT-PCBA: An Unbiased Data Set for Machine Learning and Virtual Screening. *J Chem Inf Model* 60:4263–4273. doi:10.1021/acs.jcim.0c00155 [4] Abo-Dahab Y, Xiang X, Chun J, Zhao L (2026) Benchmarking Single-Pose Docking, Consensus Rescoring, and Supervised ML on the LIT-PCBA Library: A Critical Evaluation of DiffDock, AutoDock-GPU, GNINA, and DiffDock-NMDN. arXiv:2605.01681 [5] Sunseri J, Koes DR (2021) Virtual Screening with Gnina 1.0. *Molecules* 26(23):7369. doi:10.3390/molecules26237369 [6] Furui K, Ohue M (2025) Boltzina: Efficient and Accurate Virtual Screening via Docking-Guided Binding Prediction with Boltz-2. arXiv:2508.17555 [7] Wan S, Zhang X, Xue X, Coveney PV (2026) On the Reliability of AI Methods in Drug Discovery: Evaluation of Boltz-2 for Structure and Binding Affinity Prediction. arXiv:2603.05532 [8] Huang A, Knight IS, Naprienko S (2025) Data Leakage and Redundancy in the LIT-PCBA Benchmark. arXiv:2507.21404

Supplementary material

S1. Pre-registration index. Every reading rule — endpoint, threshold, control, abort condition — was committed before the corresponding data existed. Each row gives the file, the commit that introduced it, and that commit’s date. Amendments are appended to the file rather than editing the original text, with the reason recorded in place.

Paths are relative to analysis/; the file is PREREGISTRATION.md except where named.

arm	directory	public	origin	committed
Scoring function (gnina rescore)	three_arm_docking/	f5e42306a	cd8f55639	2026-08-31 08:46
Receptor preparation repair	receptor_prep/	05d9f78ec	1f68b20c4	2026-08-31 14:26
Pose ensemble vs top pose	pose_ensemble/	d5d08c1ac	448d954e5	2026-09-02 22:52
Pose generator (DiffDock-L)	three_arm_docking/	f5e42306a	cd8f55639	2026-08-31 08:46
Boltz-2, Mpro, arm 1 (blind)	boltz2/	8df29ba21	9e3083d02	2026-09-03 08:27
Boltz-2 analysis script, written before the data	boltz2/analyse_boltz2.py	42f94969b	5afb5bec8	2026-09-03 13:51
Boltz-2, Mpro, arm 2 (pocket-conditioned)	boltz2/PREREGISTRATION_ARM2.md	064969caf	e96d4fea1	2026-09-04 11:28
Boltz-2, Factor Xa	boltz2/PREREGISTRATION_TARGETS.md	00baa6db9	68bd9b94a	2026-09-04 11:40

The public repository was produced from the origin repository with `git filter-repo`, which rewrites commit hashes but preserves author and committer dates. Both hashes are given so a reader can check either. Verify a date with `git log -1 --format=%ad --date=iso <commit>`.

Commit dates above are git metadata carried through from the project’s own history; they are not a third-party timestamp, and a reader should treat them as the authors’ record rather than as independent verification of ordering.

Bars citing a “ ≈ 0.04 measurement floor” — the pose-ensemble arm and both Boltz-2 arms — were fixed before the floor correction described in §3.1 and S3. They are read as written; see §3.2.

The search-effort comparison (exhaustiveness 4 vs 32) and the LIT-PCBA descriptor baseline were not separately pre-registered; both are reported here as such. The search-effort arm reuses two runs that pre-dated the pre-registration programme, and the LIT-PCBA baseline applies a method already fixed for the Mpro and Factor Xa panels to a third-party dataset.

S2. Per-arm experimental conditions. The six interventions are single-factor comparisons against differing baselines, not a partition of one pipeline. Four were run before the receptor defect described in §3.2 was found and repaired.

intervention	baseline AUROC	receptor	exhaustiveness	n	poses held fixed?
Scoring function (gnina CNN)	0.3992	unprotonated	4	743	yes

intervention	baseline AUROC	receptor	exhaustiveness	n	poses held fixed?
Receptor repair	0.4079	unprotonated → protonated	4	751	receptor coordinates fixed; ligands re-docked
Pose ensemble	—	unprotonated	4	751	yes (same run, 5 modes)
Pose generator (DiffDock-L)	—	unprotonated	4 (Vina arm)	751	no
Pocket conditioning (Boltz-2)	0.7913	none — co-folding from sequence	n/a	750	n/a
Eight-fold search effort	0.4079 (exh 4)	unprotonated	4 → 32	753	no

S3. Resolution floor, both variants. analysis/docking_value/floor_interval.py, FLOOR_INTERVAL.json. Paired bootstrap over compounds, 10,000 resamples, unprotonated receptor.

	value	95% CI	n	r between the two score vectors
Protocol floor — exhaustiveness-32 Vina cache vs gnina --score_only on the exhaustiveness-4 poses	0.0393	[+0.0235, +0.0557]	745	0.9372
Scoring floor — exhaustiveness-4 Vina cache vs gnina --score_only on the same exhaustiveness-4 poses	0.0201	[+0.0112, +0.0286]	743	0.9700

Neither score vector contains a compound differing from its counterpart by more than 2 kcal/mol. An earlier description of the protocol floor as a comparison of “identical poses” was incorrect and is corrected in analysis/docking_value/FLOOR_CORRECTION.md.

S4. LIT-PCBA per-target results. analysis/litpcba/descriptor_baseline_full.json and descriptor_baseline_ave.json.

target	full: actives	inactives	AUROC	AVE: actives	inactives	AUROC
ADRB2	17	25,000	0.7008	4	25,000	0.6016
ALDH1	7,168	25,000	0.6169	1,343	25,000	0.6123
ESR1_ago	13	5,583	0.7260	3	908	0.8737
ESR1_ant	102	4,948	0.7525	25	794	0.6497
FEN1	369	25,000	0.7596	91	25,000	0.7471
GBA	166	25,000	0.7456	41	25,000	0.7390
IDH1	39	25,000	0.7921	9	25,000	0.7838
KAT2A	194	25,000	0.6656	48	25,000	0.6032

target	full: actives	inactives	AUROC	AVE: actives	inactives	AUROC
MAPK1	308	25,000	0.7285	77	15,250	0.7129
MTORC1	97	25,000	0.5945	24	8,243	0.5686
OPRK1	24	25,000	0.8640	6	25,000	0.7672
PKM2	546	25,000	0.6743	136	25,000	0.6068
PPARG	27	5,211	0.8459	6	861	0.7195
TP53	79	4,168	0.5769	16	795	0.5050
VDR	884	25,000	0.6918	165	25,000	0.6510
median			0.7260			0.6510

S5. Panels. Mpro: 751 compounds, 257 active (34.2%), PDB 7VU6. Factor Xa: 886 compounds, 405 active (45.7%). The Factor Xa marginal-value comparison in §3.4 uses an earlier 854-compound freeze of the same panel; where a number from that freeze is quoted it is labelled.